PEPTIC ULCER DISEASE (PUD)

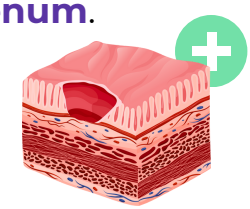
Definition

A **Peptic Ulcer** is a focal break in the gastric or duodenal mucosa.
It results from an imbalance between:

- **Defensive Factors:** Mucosal integrity, bicarbonate, and prostaglandins.
- **Aggressive Factors:** Gastric acid and Pepsin.

Key Pathological Features

- **Depth**: Must extend through the **muscularis mucosae.**
- **Size**: Typically **> 5 mm** in diameter (smaller breaks are classified as "erosions").
- **Locations**: Most common in the **stomach and proximal duodenum**. Can also occur in the lower esophagus, distal duodenum, or jejunum.



Epidemiology & Incidence

Feature	Duodenal Ulcer (DU)	Gastric Ulcer (GU)
Prevalence	**5x more common** than GU	Less common
Peak Age	30 – 55 years	55 – 70 years
Primary Site	95% in the **bulb** or pyloric channel	Gastric body or antrum

Etiology (Risk Factors)

The causes of PUD are divided into infectious, pharmacological, and physiological categories.

A. Major Causes

- **H. pylori Infection**: The leading cause of non-NSAID related ulcers.
- **NSAIDs & Aspirin**: The second most common cause. Toxicity is typically dose-related.

B. Other Medications

The following drugs are implicated in impairing mucosal defenses:

1. Corticosteroids
2. Bisphosphonates
3. Potassium Chloride (irritant effect)
4. 5-Fluorouracil (chemotherapy)

C. Lifestyle & Hypersecretory States

1. **Habits**: Smoking and excessive alcohol intake.
2. **Zollinger-Ellison Syndrome (ZES)**: A gastrin-secreting tumor causing extreme acid hypersecretion.
3. **Idiopathic**: Up to 25% (one-fourth) of ulcers have no identifiable cause (non-H. pylori, non-NSAID).

Rare Etiologies of PUD

While H. pylori and NSAIDs dominate, these secondary causes must be considered:

- **Malignancy**: Primary gastric cancer, lymphomas, or metastatic lung cancer.
- **Systemic Stress**: Acute illness (Stress ulcers), severe burns (Curling's ulcer), or head trauma (Cushing's ulcer).
- **Infections**: Viral (e.g., CMV/HSV) or granulomatous (Tuberculosis).
- **Chronic Disease**: Crohn's disease of the upper GIT.
- **Iatrogenic**: Radiation therapy or chemotherapy.
- **Vascular**: Ischemia/vascular insufficiency.

Pharmacology Note: COX-2 Inhibitors

Drugs: Celecoxib, Rofecoxib, Valdecoxib.

Enzyme	Function	Effect of Selective Inhibitors
COX-1	Synthesis of **PGE2 & PGF2α** (Gastric Cytoprotection).	**Spared**: Maintains mucosal defense.
COX-2	Mediates inflammation and pain.	**Inhibited**: Reduces systemic inflammation.

Clinical Significance: These agents were developed to provide anti-inflammatory benefits while reducing the risk of PUD compared to traditional NSAIDs.

Clinical Findings

A. Epigastric Pain (Dyspepsia)

The classic presentation of PUD is characterized by:

1. **Nature**: Gnawing, dull, aching, or "hunger-like."
2. **Food Relationship**: * Duodenal Ulcers (DU): Often relieved by food/antacids; pain recurs 2–4 hours later.
    a. Gastric Ulcers (GU): Pain may be exacerbated or unaffected by food.
3. **Nocturnal Pain**: Awakens the patient (classic for DU and 1/3 of GU cases).
4. **Periodicity**: Symptomatic weeks followed by pain-free months/years.
5. **Pain Change**: If pain becomes constant or radiating, suspect complications like penetration or perforation.

B. Associated Symptoms

1. **Nausea and Anorexia**.
2. **GIT Bleeding**: Manifests as hematemesis (bright red or "coffee ground") or melena (black, tarry stools).
3. **Obstruction Signals**: Significant vomiting and weight loss are unusual in uncomplicated PUD and suggest Gastric Outlet Obstruction or Malignancy.

"Red Flag" (Alarm) Symptoms

The presence of these signs necessitates **urgent endoscopy** to rule out malignancy or severe complications:

1. **Unintentional weight loss**
2. **Progressive dysphagia** (difficulty swallowing)
3. **Overt GI bleeding** (hematemesis or melena)
4. **Iron deficiency anemia** (suggests chronic occult blood loss)
5. **Recurrent emesis** (persistent vomiting)
6. **Family history** of upper GI malignancy



Physical Examination

- **Uncomplicated PUD**: Often normal. May show localized epigastric tenderness.
- **Complicated PUD**: Pallor/Tachycardia: Suggests anemia or acute GI bleeding.
    - **Board-like Rigidity**: Suggests perforation (peritonitis).
    - **Succussion Splash**: A "sloshing" sound heard over the stomach, suggesting Gastric Outlet Obstruction (GOO).

Investigations

A. Laboratory Findings

1. **CBC**: Microcytic hypochromic anemia (chronic blood loss); Leukocytosis (suggests penetration/perforation).
2. **Serum Amylase**: Elevated levels suggest penetration into the pancreas.
3. **Fasting Serum Gastrin**: To screen for Zollinger-Ellison Syndrome (indicated in recurrent/intractable ulcers).

B. Imaging & Endoscopy

1. **Upper Endoscopy (Gold Standard)**: Procedure of choice.
   a. Note: Duodenal ulcers are almost never malignant (no biopsy needed). Gastric ulcers MUST be biopsied to rule out cancer.
2. **Barium Study**: Limited accuracy compared to endoscopy.
3. **CT Abdomen with Contrast**: Best for identifying complications (perforation, GOO) rather than the ulcer itself.
4. **Plain X-ray (Erect)**: Shows air under the diaphragm in cases of perforation.

Management of PUD

I. General Measures

1. **Stress reduction and regular healthy meals.**
2. **Cessation of smoking and alcohol.**
3. **Discontinuation of NSAIDs.**



II. Pharmacotherapy

Class	Examples	Mechanism / Notes
Proton Pump Inhibitors (PPIs)	Omeprazole, Esomeprazole, Pantoprazole	Drug of choice. Binds H+/K+ ATPase. Inhibits >90% of acid. Take 30 mins before breakfast.
H2 Receptor Antagonists	Ranitidine, Famotidine	Effective for nocturnal acid, less so for meal-stimulated acid. Used for uncomplicated cases.
Mucosal Protective Agents	Sucralfate, Bismuth, Misoprostol	Enhances defenses. Misoprostol is a PG analog specific for preventing NSAID-induced ulcers.

III. H. Pylori eradication

IV. Surgery

Surgical Indications

1. **Complications** (Bleeding, perforation, obstruction).
2. **Suspicion of malignancy**.
3. **Medical Failure**: Persistent symptoms despite H. pylori eradication and stopping NSAIDs.

Complications (The "Big Four")

IMPORTANT

1. Hemorrhage (Most Common)

- Presentation: Hematemesis, melena, or "coffee ground" emesis. Can lead to shock.
- Management: ICU admission, IV fluids/blood, IV PPIs, and urgent Endoscopy for hemostasis.

2. Perforation

- Presentation: Sudden, severe pain; "board-like" abdominal rigidity; shock.
- Diagnosis: Air under diaphragm on erect X-ray.
- Management: Nasogastric (NG) suction, IV antibiotics, and Surgical closure with omental patch.

3. Penetration

- Definition: Ulcer erodes into adjacent organs (usually the pancreas).
- Presentation: Pain loses its rhythm, becomes constant, and radiates to the back. May cause pancreatitis.

4. Gastric Outlet Obstruction (GOO)

- Presentation: Epigastric fullness, vomiting of foul-smelling undigested food, weight loss, and alkalosis.
- Causes: Edema (acute) or scarring/fibrosis (chronic).
- Management: NG suction (1 week), IV fluids, PPIs. Chronic cases require Endoscopic balloon dilatation or surgery.

Malignant Transformation

- **Gastric Ulcers**: Higher risk of malignancy. Always require follow-up endoscopy to ensure healing and biopsies to rule out adenocarcinoma.
- Role of H. pylori: Increases gastric cancer risk 3–6 times.

PEPTIC ULCER DISEASE

Titles

- Definition
- Pathology
- Epidemiology
- Etiology
- Pharmacology
- Clinical picture
- Investigations
- Management
- Complications

Your guide to master this book

Abdurrahman Hafez

Titles of the topic

Titles

- Definition
- LM
- EM
- Molecular structure
- Functions

Remember

A point mentioned in other topic in the book and may cause some confusion

Remember

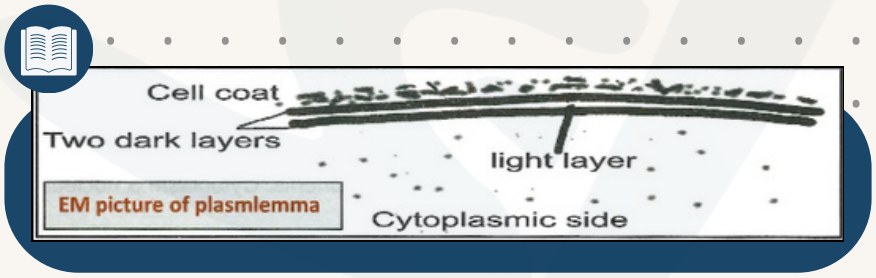
Mnemonics

Mnemonic

Important

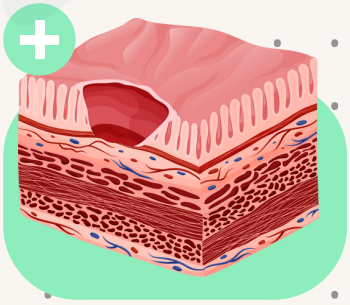
IMPORTANT

Images from the handout



Additional images for Understanding





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